Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

Section A **3*10=30**

1)	State the definition of a simple graph. OR Define a directed graph and provide an example.	**3**	(CO1, L2)
2)	What is the degree of a vertex in graph theory? Explain with an example.	**3**	(CO1, L2)
3)	Explain the concept of a Hamiltonian path in graph theory. OR Describe what is a Eulerian circuit and give an example.	**3**	(CO1, L2)
4)	Define the chromatic number of a graph. OR What is an independent set in a graph and give an example.	**3**	(CO1, L2)
5)	What does the Handshaking Lemma state in graph theory? OR Explain what is a spanning tree and its significance.	**3**	(CO1, L2)
6)	Describe the process of finding a shortest path in a weighted graph using Dijkstra's algorithm. OR Explain how to use Depth-First Search (DFS) for cycle detection.	**3**	(CO2, L4)
7)	Prove that a complete graph with n vertices has (n-1) edges. OR Prove that in any graph, the sum of the degrees of all vertices is twice the number of edges.	**3**	(CO2, L4)
8)	Using Kruskal's algorithm, find a minimum spanning tree for the given graph. OR Use Prim’s algorithm to construct a minimum spanning tree.	**3**	(CO2, L4)
9)	Construct an adjacency matrix for the given graph. OR Draw a directed graph from its incidence list.	**3**	(CO1, L2)
10)	Describe how to use the Hungarian algorithm for solving assignment problems. OR Explain the Floyd-Warshall algorithm for finding all-pairs shortest paths.	**3**	(CO2, L4)

Section B **7*3=21**

1) a)	Prove that in any graph, the number of vertices with odd degree is always even.	**7**	(CO2, L4)
	OR		
b)	Explain the concept of a bipartite graph and provide an example.	**7**	(CO1, L2)

2) a)	Prove that a graph is connected if and only if there exists an Eulerian circuit.	**7**	(CO2, L4)
	OR		
b)	Describe the process of edge coloring in a graph and provide an example.	**7**	(CO1, L3)
3) a)	Prove that if G is a connected graph with n vertices, then the number of edges in its spanning tree is (n-1).	**7**	(CO2, L4)
	OR		
b)	Explain the concept of a Hamiltonian cycle and provide an example.	**7**	(CO1, L2)

Section C **7*4=28**

1) a)	Given a weighted graph, apply Dijkstra's algorithm to find the shortest path from vertex A to all other vertices.	**7**	(CO2, L4)
	OR		
b)	Design a minimum spanning tree for the given graph using Prim's algorithm.	**7**	(CO2, L4)
2) a)	Given a graph, determine if it contains an Eulerian path and justify your answer.	**7**	(CO1, L3)
	OR		
b)	Prove that a graph has an Eulerian circuit if and only if every vertex has even degree.	**7**	(CO2, L4)
3) a)	Explain the concept of a perfect matching in bipartite graphs and provide an example.	**7**	(CO1, L2)
	OR		
b)	Using the Hungarian algorithm, find a minimum cost perfect matching in a given bipartite graph.	**7**	(CO2, L4)

Section D **7*3=21**

1) a)	Apply Prim's algorithm to find the minimum spanning tree for a given network and calculate its total weight.	**7**	(CO2, L4)
	OR		
b)	Describe the process of edge coloring a graph and find the chromatic number for a given graph.	**7**	(CO2, L4)
2) a)	Given a graph, determine if it is planar and justify your answer using Kuratowski's theorem.	**7**	(CO1, L3)
	OR		
b)	Explain the concept of a graph isomorphism and provide an example.	**7**	(CO1, L3)
3) a)	Using the Floyd-Warshall algorithm, find all-pairs shortest paths in a given weighted graph.	**7**	(CO2, L4)
	OR		
b)	Explain the concept of a vertex cover and provide an example.	**7**	(CO1, L2)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

AI-assisted draft - teacher review and approval required before use